CLASSIFICATION AND DETECTION OF OCT RETINAL IMAGE USING DEEP LEARNING

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is a Bonafide record of the work done by the above-mentioned candidates in partial fulfilment of the requirements for the award of the degree of **"BACHELOR OF COMPUTER APPLICATIONS"** and has been carried out under my guidance during the Sixth semester of the academic year (2025–2026).

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Viva – Voce Examinations

The Project Viva-Voce Examinations was held on ----------

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Signature of External Examiner

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ABSTRACT

OCT (Optical Coherence Tomography) is a special eye-scan that takes detailed cross-section images of the retina. Doctors use these images to find eye diseases like Diabetic Retinopathy, Macular Edema, Age related Macular Degeneration (AMD), etc. Manually checking every OCT image takes a lot of time and may lead to human error.

In this project, we build a deep learning-based system that can automatically detect and classify retinal diseases from OCT images. Deep learning models like CNN (Convolutional Neural Network) learn patterns from thousands of eye images. These models can identify even very small changes in the retina that may not be easily seen by humans.

CHAPTER-1

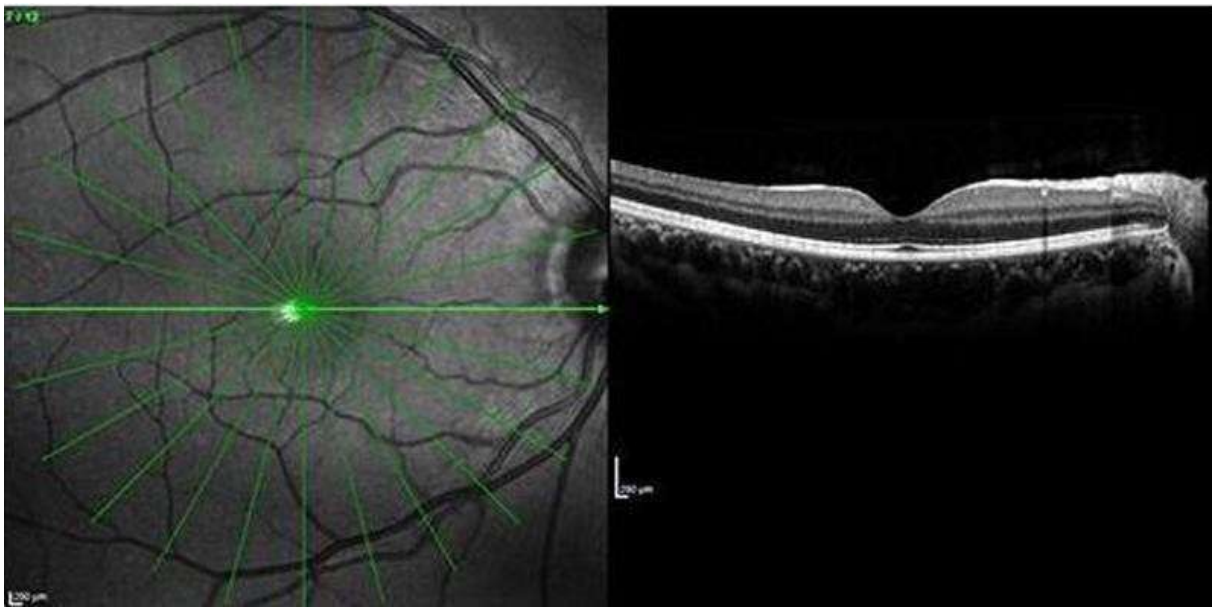
INTRODUCTION

The macular area is an important area of the retina, which is related to fine vision, color vision, and other visual functions. The vision will be negatively affected once lesions occur in the macular area. Senile macular degeneration, also known as age-related macular degeneration (AMD), is the aging change in the macular area, which occurs mainly in people over 45 years old and the prevalence rate of AMD increases with age, making it one of the primary causes of blindness in the elderly. Diabetic macular edema (DME) is a major complication of diabetes in the eyes, which is the leading cause of blindness in young adults in developed countries. Therefore, early detection is important for the treatment of AMD and DME.

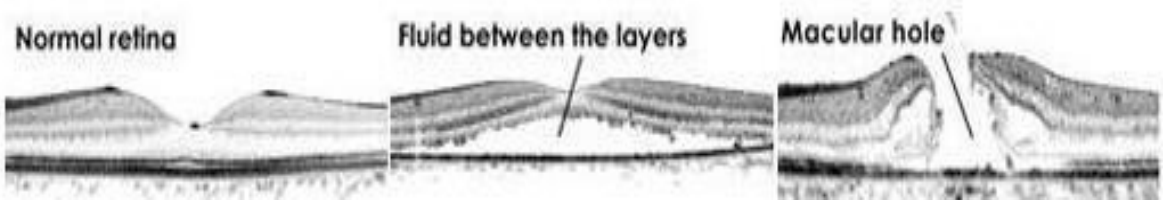
OCT is an imaging technique that detects the back reflection of the incoming weak coherent light or several scattered signal in different levels of the biological tissue based on the fundamental of weak coherence light interference, thus the two or three dimensional structural image of the biological tissue can be obtained through scanning. OCT is widely used in the diagnosis of AMD and DME.

Automated detection and diagnosis of retinopathy can assist clinicians in diagnosis and treatment, which is conducive to the early detection and treatment of patients, saves medical resources and helps to reduce the cost of treatment of patients. In addition, the automatic analysis of medical images through the existing image analysis and processing technologies has become an important research field. There are two problems in retinal OCT images analysis. First, the retinal OCT image public dataset used for research is smaller in number scale compared with the dataset in the fields of traditional image analysis. Second, there are huge differences in the quality and size of retinal OCT the images collected in different tissues (huge differences also exist in the retinal OCT images encountered by doctors in clinical medicine).

1.1 OPTICAL COHERENCE TOMOGRAPHY



Optical Coherence Tomography (OCT) is a non-invasive, non-contact imaging system which provides high resolution cross-sectional images of the retina, vitreous and optic nerve. OCT is frequently used to measure retinal thickness which guides treatment decisions. It allows detection of very fine retinal structures and pathology, sometimes unidentifiable on clinical exam. It is widely used in the diagnosis of macular diseases, including age-related macular degeneration, diabetic macular edema, macular hole, epiretinal membrane, vitreomacular traction and central serous chorioretinopathy, to name just a few. Newer generation OCT machines are very fast and allow for test completion in a couple of minutes with no discomfort for the patient. The layers within the retina can be differentiated and retinal thickness can be measured to aid in the early detection and diagnosis of retinal diseases and conditions.



OCT testing has become a standard of care for the assessment and treatment of most retinal conditions. OCT uses rays of light to measure retinal thickness. No radiation or X-rays are used in this test, an OCT scan does not hurt and it is not uncomfortable. You may be given an OCT scan for a variety of reasons, including monitoring of the progress of your disease, verifying or discounting suspected swelling of the retina or checking OCT results against other results to determine the effectiveness of the current medication regime. Optical Coherence Tomography uses technology that is best compared to ultrasound, except that it employs light rather than sound and thereby achieves clearer, sharper resolution.

1.2 DEEP LEARNING

Deep learning (also learning data known as deep structured learning or hierarchical learning) is part of a broader family of machine learning methods based on representations, as opposed to task-specific algorithms. Learning can be supervised, semi-supervised or unsupervised. Detecting fire in images using image processing and computer vision techniques has gained a lot of attention from researchers during the past few years. Indeed, with sufficient accuracy, such systems may outperform traditional fire detection equipment. One of the most promising techniques used in this area is Convolutional Neural Networks (CNNs). However, the previous research on fire detection with CNNs has only been evaluated on balanced datasets, which may give misleading information on real-world performance, where fire is a rare event. Actually, as demonstrated in this paper, it turns out that a traditional CNN performs relatively poorly when evaluated on the more realistically balanced benchmark dataset provided in this paper. We there fore propose to use even deeper Convolutional Neural Networks for fire detection in images, and enhancing these with fine tuning based on a fully connected layer. We use two pre-trained state-of-the-art Deep CNNs, VGG16 and Resnet50, to develop our fire detection system. The Deep CNNs are tested on our imbalanced dataset, which we have assembled to replicate real world scenarios. It includes images that are particularly difficult to classify and that are deliberately unbalanced by including significantly more non-fire images than fire images. The dataset has been made available online. Our results show that adding fully connected layers for fine tuning indeed does increase accuracy, however, this also increases training time.

It allows us to train an AI to predict outputs, given a set of inputs. Both supervised and unsupervised learning can be used to train the AI. We will learn how deep learning works by building an hypothetical airplane ticket price estimation service. The iterative aspect of machine learning is important because as models are exposed to new data, they are able to independently adapt. A convolutional neural network (CNN, or ConvNet) is a class of deep neural networks, most commonly applied to analyzing visual imagery Individual cortical neurons respond to stimuli only in a restricted region of the visual field known as the receptive field.

1.2.1 APPLICATIONS

- Deep Learning is being widely used in industries to solve large number of problems like computer vision,
- Natural language processing and pattern recognition.
- Image Recognition: Deep Neural Nets are used to identify objects in an image.
- Speech recognition in voice assistants like *Amazon Alexa*.
- Self-driving car technology developed by *Tesla*.
- Medical image analysis for disease detection.
- Fraud detection in banking and financial transactions.
- Recommendation systems used by *Netflix*.
- Robotics and industrial automation.
- Text generation systems like *ChatGPT*.
- Cybersecurity for detecting malware and spam emails.

1.3 MACHINE LEARNING

Predictive analytics tools are powered by several different models and algorithms that can be applied to wide range of use cases. Determining what predictive modeling techniques are best for your company is key to getting the most out of a predictive analytics solution and leveraging data to make insightful decisions. In the statistical context, Machine Learning is defined as an application of artificial intelligence where available information is used through algorithms to process or assist the processing of statistical data. While Machine Learning

involves concepts of automation, it requires human guidance. Machine Learning involves a high level of generalization in order to get a system that performs well on yet unseen data instances Machine learning is a relatively new discipline within Computer Science that provides a collection of data analysis techniques. Some of these techniques are based on well-established statistical methods (e.g. logistic regression and principal component analysis) while many others are not.

Most statistical techniques follow the paradigm of determining a particular probabilistic model that best describes observed data among a class of related models. Similarly, most machine learning techniques are designed to find models that best fit data (i.e. they solve certain optimization problems), except that these machine learning models are no longer restricted to probabilistic ones.

Therefore, an advantage of machine learning techniques over statistical ones is that the latter require underlying probabilistic models while the former do not. Even though some machine learning techniques use probabilistic models, the classical statistical techniques are most often too stringent for the oncoming Big Data era, because data sources are increasingly complex and multi-faceted. Prescribingprobabilistic models relating variables from disparate data sources that are plausible and amenable to statistical analysis might be extremely difficult if not impossible.

Machine learning might be able to provide a broader class of more flexible alternative analysis methods better suited to modern sources of data. It is imperative for statistical agencies to explore the possible use of machine learning techniques to determine whether their future needs might be better met with such techniques than with traditional ones.

1.3.1 CLASSES OF MACHINE LEARNING

Examples of supervised learning

Logistic regression, when used for prediction purposes, is an example of supervised machine learning. In logistic regression, the values of a binary response variable (with values 0 or 1, say) as well as a number of predictor variables (covariates) are observed for a number of observation units. These are called training data in machine learning terminology. The main hypotheses are that the response variable follows a Bernoulli distribution (a class of probabilistic models), and the link between the response and predictor variables is the relation that the logarithm of the posterior odds of the response is a linear function of the predictors. The response variables of the units are assumed to be independent of each other, and the method

of maximum likelihood is applied to their joint probability distribution to find the optimal values for the coefficients (these parameterize the aforementioned joint distribution) in this linear function. The particular model with these optimal coefficient values is called the "fitted model," and can be used to "predict" the value of the response variable for a new unit (or, "classify" the new unit as 0 or 1) for which only the predictor values are known. Support Vector Machines (SVM) are an example of a non-statistical supervised machine learning technique; it has the same goal as the logistic regression classifier just described:

Given training data, find the best-fitting SVM model, and then use the fitted SVM model to classify

1.3.2 The most widely used predictive models are

Decision trees

Decision trees are a simple, but powerful form of multiple variable analysis. They are produced by algorithms that identify various ways of splitting data into branch-like segments. Decision trees partition data into subsets based on categories of input variables, helping you to understand someone's path of decisions.

Regression (linearandlogistic)

Regression is one of the most popular methods in statistics. Regression analysis estimates relationships among variables, finding key patterns in large and diverse data sets and how they relate to each other.

Neural networks

Patterned after the operation of neurons in the human brain, neural networks (also called artificial neural networks) are a variety of deep learning technologies. They're typically used to solve complex pattern recognition problems – and are incredibly useful for analyzing large data sets. They are great at handling nonlinear relationships in data – and work well when certain variables are unknown

Developing the right environment

While machine learning and predictive analytics can be a boon for any organization, implementing these solutions haphazardly, without considering how they will fit into everyday operations, will drastically hinder their ability to deliver the insights the organization needs

the organization's needs. Today, however, predictive analytics and machine learning is no longer just the domain of mathematicians, statisticians and data scientists, but also that of business analysts and consultants. More and more of a business' employees are using it to develop insights and improve business operations – but problems arise when employees do not know what model to use, how to deploy it, or need information right away.

At SAS, we develop sophisticated software to support organizations with their data governance and analytics. Our data governance solutions help organizations to maintain high-quality data, as well as align operations across the business and pinpoint data problems within the same environment. Our predictive analytics solutions help organizations to turn their data into timely insights for better, faster decision making. These predictive analytics solutions are designed to meet the needs of all types of users and enables them to deploy predictive models rapidly.

1.4 IMAGE PROCESSING FUNDAMENTAL

Digital image processing refers processing of the image in digital form. Modern cameras may directly take the image in digital form but generally images are originated in optical form. They are captured by video cameras and digitalized. The digitalization process includes sampling, quantization. Then these images are processed by the five fundamental processes, at least any one of them, not necessarily all of them.

1.4.1 IMAGE PROCESSING TECHNIQUES

This section gives various image processing techniques.

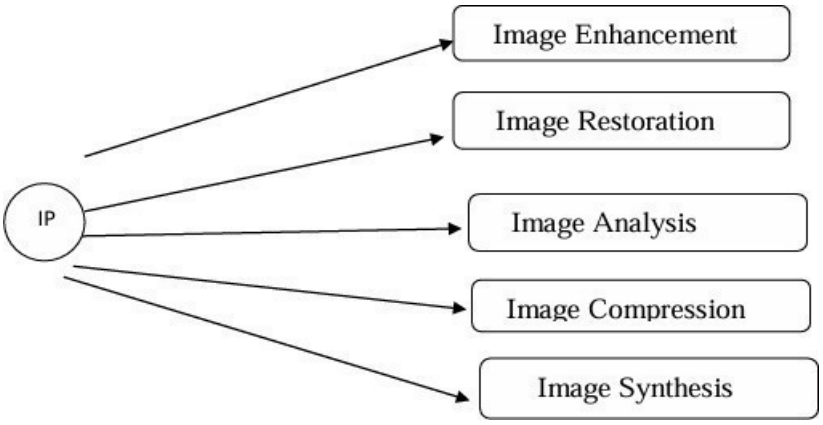


FIG 1.3: IMAGE PROCESSING TECHNIQUES

IMAGE ENHANCEMENT

Image enhancement operations improve the qualities of an image like improving the image's contrast and brightness characteristics, reducing its noise content, or sharpen the details. This just enhances the image and reveals the same information in more understandable image. It does not add any information to it.

IMAGE RESTORATION

Image restoration like enhancement improves the qualities of image but all the operations are mainly based on known, measured, or degradations of the original image. Image restorations are used to restore images with problems such as geometric distortion, improper focus, repetitive noise, and camera motion. It is used to correct images for known degradations.

IMAGE ANALYSIS

Image enhancement operations improve the qualities of an image like improving the image's contrast and brightness characteristics, reducing its noise content, or sharpen the details. This just enhances the image and reveals the same information in more understandable image. It does not add any information to it.

IMAGE COMPRESSION

Image compression and decompression reduce the data content necessary to describe the image. Most of the images contain lot of redundant information, compression removes all the redundancies. Because of the compression the size is reduced, so efficiently stored or transported. The compressed image is decompressed when displayed. Lossless compression preserves the exact data in the original image, but lossy compression does not represent the original image but provide excellent compression.

IMAGE SYNTHESIS

Image synthesis operations create images from other images or non-image data. Image synthesis operations generally create images that are either physically impossible or impractical to acquire.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

In existing system OCT image segmentation using machine learning and supervised learning algorithm sequential minimal optimization scheme has been has been implemented in exist model implemented. CNN is of the well-regarded machine learning method sin the literature. One of the reasons of its popularity is due to the automatic hierarchical feature representation in recognizing objects and patters in images CNNs reduce the parameters of a given problem using spatial relationships between them. This makes them a more practical classifier specially in image processing where we deal with a large number of parameters (pixels), rotation, translation, and scale of images. In fact, CNNs alleviate the drawbacks of Feel Forward Neural networks and Multi-Layer Perceptions by using an alternative to matrix multiplication. We use this powerful method in this study due to the nature of OCT Image Classification via Deep Learning.

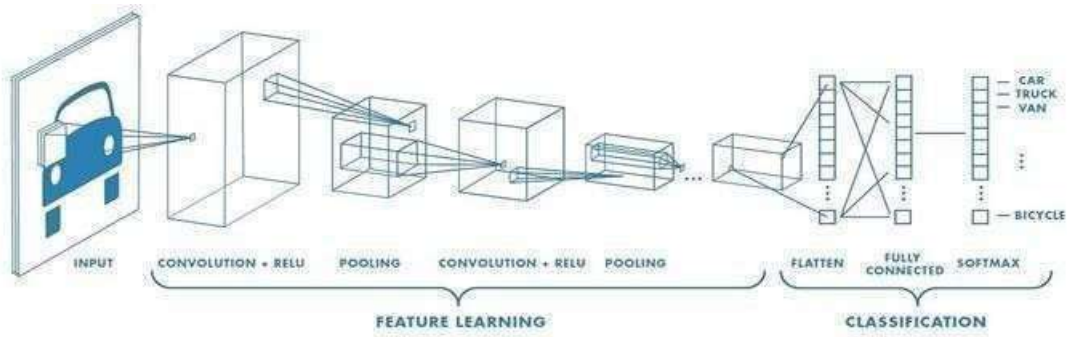


Figure: Retinal OCT (optical coherence tomography) Images classification using CNN

DISADVANTAGE

- They do not take into consideration of the noise or the image normalization in the input retinal image.
- These methods does not show high contrast image for the output image.
- Manual assessment is subjective, time consuming and expensive.
- Poor and Inaccurate segmentation results
- All Optic disc like regions are misclassified since it is hard to distinguish from fragments of the vascular system or from certain eye feature.

2.2 PROPOSED SYSTEM

The proposed system oct image classification using deep learning algorithm Deep learning convolution neural network based algorithm is implemented for extract the oct image feature The features are extracted and stored into the data base OCT image features are extracted and trained using the deep neural network.

The public OCT dataset proposed to classify and recognize OCT images. They fine-tuned the pre-trained convolution neural network (CNN) GoogleLeNet to train and classify OCT images so that the neural network can be trained and good results can be achieved by using limited data. Also used transfer learning to classify OCT images in order to reduce the dependence on dataset size. They fine-tuned and trained the pre-trained VGG1.

To conduct the classification analysis of OCT images and performed recognition classification by employing the linear configuration pattern (LCP) feature of OCT image and sequential minimal optimization (SMO) algorithm. It is worth mentioning that multi-scale method was used in the extraction of the feature so that the LCP features can be calculated on multiple OCT scales, which makes the local information and the overall information of the OCT image complementary in the extracted features.

ADVANTAGE

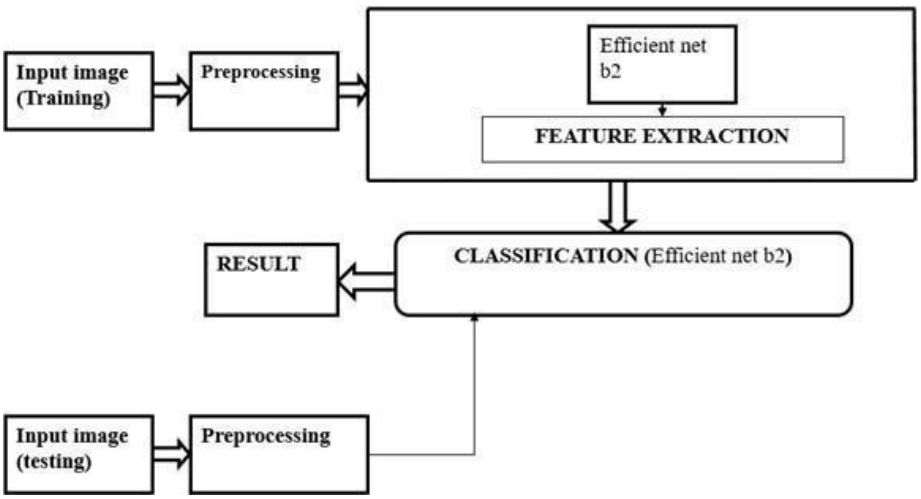
- Nearly Accurate Detection of Glaucoma compared to existing approaches.
- The noise content present in the retinal image is removed properly
- Segmentation results of Optic disc and Optic cup regions are better.
- The input retinal image is properly classified as normal or moderate or severe based on CDR ratio.

2.3 PROPOSED METHOD

We propose to use the characteristics of the network structure to reduce the dependence on the size of the dataset and enhance the adaptability to the differences among different datasets. Some network models with effective features of reusable features were selected and trained. For the selected network models, corresponding comparative evaluation was conducted and transfer learning has been achieved for the network with the pre-training model. And the parameters of each network model was tuned to make it achieve better results. In order to verify and compare the advantages and disadvantages of the network model with reusability function with the traditional network model and method, the transfer learning of VGG16, VGG19, Inception-v3 and the classification method using SVM were realized. The models involved in the training evaluation were as follows: Clique Net, DPN92, DenseNet 121, ResNet 50, and ResNext101.

2.3 SYSTEM ARCHITECTURE:

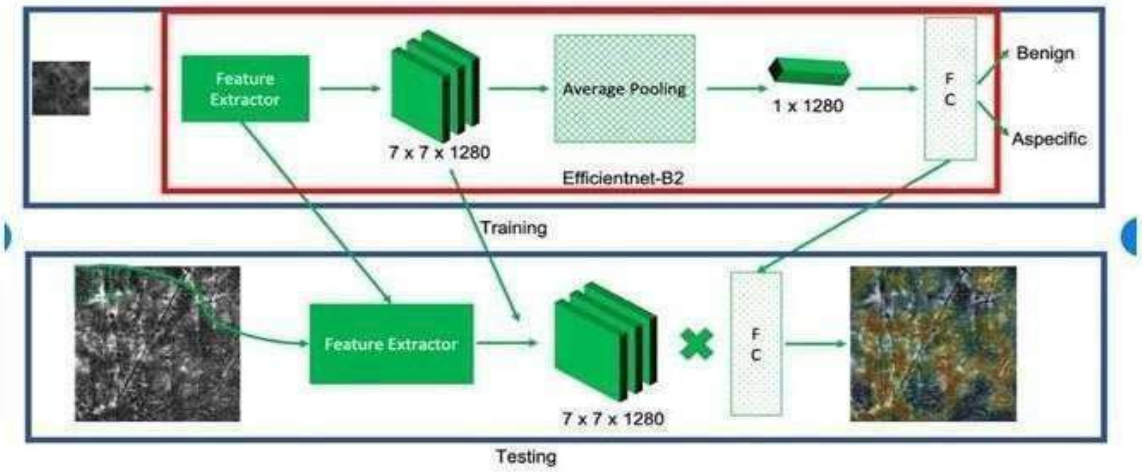
A system architecture is the conceptual model that defines the structure, behavior, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures and behaviors of the system.



2.4 ALGORITHM

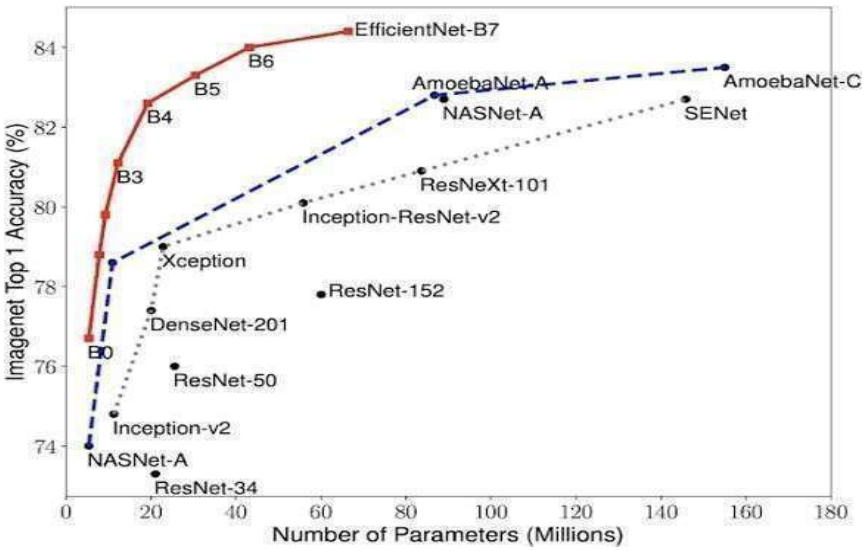
EFFICIENT NET B2:

Efficient Net is a convolutional neural network architecture and scaling method that uniformly scales all dimensions of depth/width/resolution using a compound coefficient.



An overview of the algorithm—Efficientnet-B2 is trained for classification in the standard manner (note that FC denotes the fully connected final layer). In testing, the mosaic is broken up into patches that are fed to the feature extractor. The extracted features are then multiplied by the weights in the fully connected layer to create a CAM.

The authors wanted to optimize for accuracy and efficiency. So, they performed a neural architecture search. This search yielded the Efficient-B0 architecture which looks pretty simple and straightforward to implement.



Accuracy Comparison. EfficientNet-B0 is the baseline network developed by AutoML MNAS, while Efficient-B1 to B7 are obtained by scaling up the baseline network. In particular, our EfficientNet-B7 achieves new state-of-the-art 84.4% top 1 / 97.1% top-5 accuracy, while being 8.4x smaller than the best existing CNN.

CHAPTER 3

MODULE IMPLEMENTATION

3.1 MODULE LIST

- Dataset collection
- Preprocessing
- Model development
- Performance evaluation

3.2 MODULE DESCRIPTION

3.2.1 DATASET COLLECTION

- In this process to collect a data from Kaggle. This Kaggle data collection has an only numerical value and in this data using multiple purpose.
- Kaggle supports a variety of dataset publication formats, but we strongly encourage dataset publishers to share their data in an accessible, non proprietary format if possible.
- Not only are open, accessible data formats better supported on the platform, they are also easier to work with for more people regardless of their tools.

3.2.2 PREPROCESSING

- Data preprocessing is a process of preparing the raw data and making it suitable for a deep learning model.
- It is the first and crucial step while creating a deep learning model. When creating a machine learning project, it is not always a case that we come across the clean and formatted data. And while doing any operation with data, it is mandatory to clean it and put in a formatted way. So for this, we use data preprocessing task.

- A real-world data generally contains noises, missing values, and maybe in an unusable format which cannot be directly used for machine learning models.

- Data preprocessing is required tasks for cleaning the data and making it suitable for a machine learning model which also increases the accuracy and efficiency of a machine learning mode.

3.2.3 MODEL DEVELOPMENT

- The model maintenance plays critical role once the model is deployed into production.

- The maintenance of machine learning model includes keeping the model up to date and relevant in tune with the source data changes as there is a risk of model becoming outdated in course of time. Also, the configuration management of ML model play an important role in model management as the number of models grow.

- This article focuses on principles and industry standard practices, including the tools and technologies used for ML model development, deployment, and maintenance in an enterprise environment.

3.2.4 PERFORMANCE EVALUATION

- In this process to implement of project Accuracy, recall, Precision, FIS core and ROC curve.

- Percentage of positive instances out of the total predicted positive instances. Here denominator is the model prediction done as positive from the whole given dataset. Take it as to find out 'how much the model is right when it says it is right'.

CHAPTER 4

SYSTEM SPECIFICATION

H/W SYSTEM CONFIGURATION

• processor - Pentium – IV

• RAM - 4 GB (min)

• Hard Disk - 20 GB

S/W SYSTEM CONFIGURATION

• Operating System : Windows 7 or 8

• Software : python Idle

SOFTWARE ENVIRONMENT

Python Technology

Python is an interpreter, high-level, general-purpose programming language. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming. Python is often described as a "batteries included" language due to its comprehensive standard library.

Python Programing Language

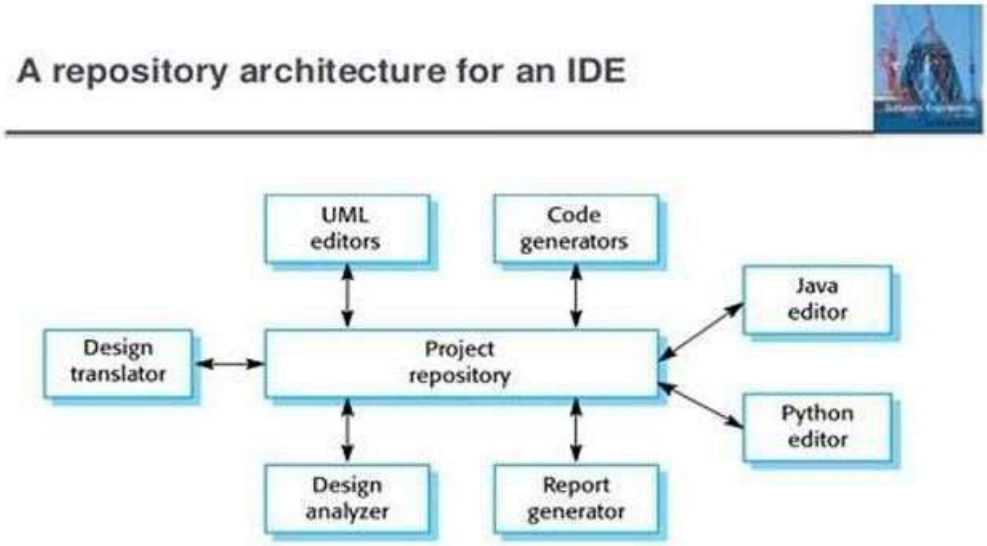
Python is a multi-paradigm programming language. Object-oriented programming and structured programming are fully supported, and many of its features support functional programming and aspect-oriented programming (including by Meta programming and met objects (magic methods)). Many other 29 paradigms are supported via extensions, including design by contract and logic programming.

Python packages with a wide range of functionality, including:

- Easy to Learn and Use
- Expressive Language
- Interpreted Language
- Cross-platform Language

- Free and Open Source
- Object-Oriented Language
- Extensible
- Large Standard Library
- GUI Programming Support

Integrated Python uses dynamic typing and a combination of reference counting and a cycle detecting garbage collector for memory management. It also features dynamic name resolution (late binding), which binds method and variable names during program execution. Rather than having all of its functionality built into its core, Python was designed to be highly extensible. This compact modularity has made it particularly popular as a means of adding programmable interfaces to existing applications. Van Rossum's vision of a small core language with a large standard library and easily extensible interpreter stemmed from his frustrations with ABC, which espoused the opposite approach. Python is meant to be an easily readable language. Its formatting is visually uncluttered, and it often uses English keywords where other languages use punctuation. Unlike many other languages, it does not use curly brackets to delimit



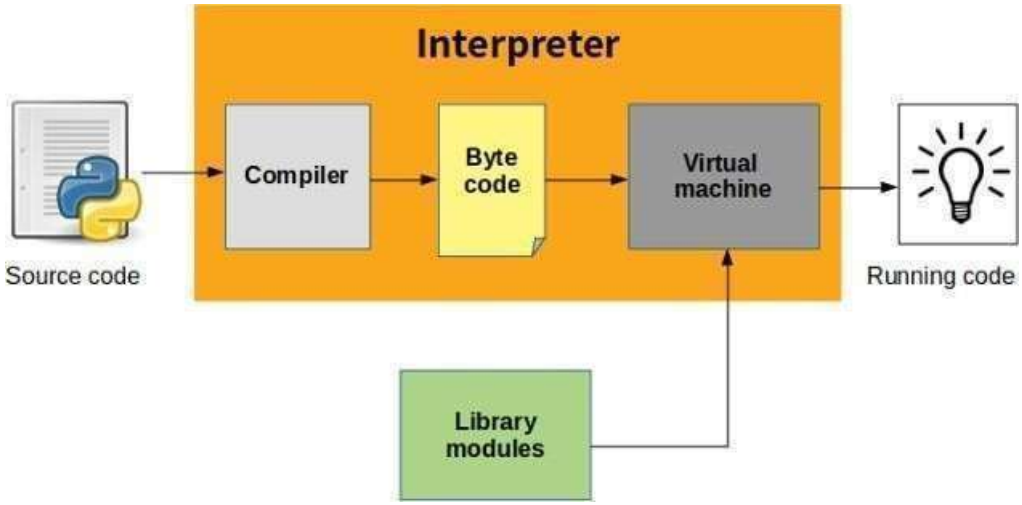
blocks, and semicolons after statements are optional. It has fewer syntactic exceptions and special cases than C or Pascal.

Python strives for a simpler, less-cluttered syntax and grammar while giving developers a choice in their coding methodology. In contrast to Perl's "there is more than one way to do it" motto, Python embraces a "there should be one and preferably only one obvious way to do it" design philosophy. Alex Martelli, a Fellow at the Python Software Foundation and Python book author, writes that "To describe something as 'clever' is not considered a compliment in the Python culture."

Python's developers strive to avoid premature optimization, and reject patches to non-critical parts of the Python reference implementation that would offer marginal increases in speed at the cost of clarity. When speed is important, a Python programmer can move time-critical functions to extension modules written in languages such as C, or use PyPy, a just-in-time compiler. Python is also available, which translates a Python script into C and makes direct C-level API calls into the Python interpreter.

An important goal of Python's developers is keeping it fun to use. This is reflected in the language's name a tribute to the British comedy group Monty Python and in occasionally playful approaches to tutorials and reference materials, such as examples that refer to spam and eggs (from a famous Monty Python sketch) instead of the standard foo and bar.

Python uses duck typing and has typed objects but untyped variable names. Type constraints are not checked at compile time; rather, operations on an object may fail, signifying that the given object is not of a suitable type. Despite being dynamically typed, Python is strongly typed, forbidding operations that are not well defined (for example, adding a number to a string) rather than silently attempting to make sense of them.



The Python Platform:

The platform module in Python is used to access the underlying platform's data, such as, hardware, operating system, and interpreter version information. The platform module includes tools to see the platform's hardware, operating system, and interpreter version information where the program is running.

There are four functions for getting information about the current Python interpreter. python_version() and python_version_tuple() return different forms of the interpreter version with major, minor, and patch level components. python_compiler() reports on the compiler used to build the interpreter.

And python_build() gives a version string for the build of the interpreter. Platform() returns string containing a general purpose platform identifier. The function accepts two optional Boolean arguments. If aliased is true, the names in the return value are converted from a formal name to their more common form. When terse is true, returns a minimal value with some parts dropped.

What does python technology do?

Python is quite popular among programmers, but the practice shows that business owners are also Python development believers and for good reason. Software developers love it for its straightforward syntax and reputation as one of the easiest programming languages to learn. Business owners or CTOs appreciate the fact that there’s a framework for pretty much anything – from web apps to machine learning.

Moreover, it is not just a language but more a technology platform that has come together through a gigantic collaboration from thousands of individual professional developers forming a huge and peculiar community of aficionados.

So what are the tangible benefits the language brings to those who decided to use it as a core technology? Below you will find just some of those reasons.

PRODUCTIVITY AND SPEED

It is a widespread theory within development circles that developing Python applications is approximately up to 10 times faster than developing the same application in Java or C/C++. The impressive benefit in terms of time saving can be explained by the clean object-oriented design, enhanced process control capabilities, and strong integration and text processing capacities. Moreover, its own unit testing framework contributes substantially to its speed and productivity.

PYTHON IS POPULAR FOR WEB APPS

Web development shows no signs of slowing down, so technologies for rapid and productive web development still prevail within the market. Along with JavaScript and Ruby, Python, with its most popular web framework Django, has great support for building web apps and is rather popular within the web development community.

OPEN-SOURCE AND FRIENDLY COMMUNITY

As stated on the official website, it is developed under an OSI-approved open source license, making it freely usable and distributable. Additionally, the development is driven by the community, actively participating and organizing conference, meet-ups, hackathons, etc. fostering friendliness and knowledge sharing.

PYTHON IS QUICK TO LEARN

It is said that the language is relatively simple so you can get pretty quick results without actually wasting too much time on constant improvements and digging into the complex engineering insights of the technology. Even though Python programmers are really in high demand these days, its friendliness and attractiveness only help to increase number of those eager to master this programming language.

BROAD APPLICATION

It is used for the broadest spectrum of activities and applications for nearly all possible industries. It ranges from simple automation tasks to gaming, web development, and even complex enterprise systems.

These are the areas where this technology is still the king with no or little competence:

- Machine learning as it has a plethora of libraries implementing machine learning algorithms.
- Web development as it provides back end for a website or an app.
- Cloud computing as Python is also known to be among one of the most popular cloud-enabled languages even used by Google in numerous enterprise-level software apps.
- Scripting.
- Desktop GUI applications.

Python compiler

The Python compiler package is a tool for analyzing Python source code and generating Python bytecode. The compiler contains libraries to generate an abstract syntax tree from Python source code and to generate Python bytecode from the tree.

The compiler package is a Python source to bytecode translator written in Python. It uses the built-in parser and standard parser module to generate a concrete syntax tree. This tree is used to generate an abstract syntax tree (AST) and then Python bytecode.

The full functionality of the package duplicates the built-in compiler provided with the Python interpreter. It is intended to match its behavior almost exactly. Why implement another compiler that does the same thing? The package is useful for a variety of purposes. It can be modified more easily than the built-in compiler. The AST it generates is useful for analyzing Python source code.

The basic interface

The top-level of the package defines four functions. If you import compiler, you will get these functions and a collection of modules contained in the package.

compiler.parse(buf)

Returns an abstract syntax tree for the Python source code in buf. The function raises Syntax Error if there is an error in the source code. The return value is a compiler.ast. Module instance that contains the tree.

compiler.parseFile(path)

Return an abstract syntax tree for the Python source code in the file specified by path. It is equivalent to parse(open(path).read()).

LIMITATIONS

There are some problems with the error checking of the compiler package. The interpreter detects syntax errors in two distinct phases. One set of errors is detected by the interpreter's parser, the other set by the compiler. The compiler package relies on the interpreter's parser, so it get the first phases of error checking for free. It implements the second phase itself, and that implementation is incomplete. For example, the compiler package does not raise an error if a name appears more than once in an argument list: def f(x, x): ...

A future version of the compiler should fix these problems.

PYTHON ABSTRACT SYNTAX

The compiler.ast module defines an abstract syntax for Python. In the abstract syntax tree, each node represents a syntactic construct. The root of the tree is Module object.

The abstract syntax offers a higher level interface to parsed Python source code. The parser module and the compiler written in C for the Python interpreter use a concrete syntax tree. The concrete syntax is tied closely to the grammar description used for the Python parser. Instead of a single node for a construct, there are often several levels of nested nodes that are introduced by Python's precedence rules.

The abstract syntax tree is created by the compiler.transformer module. The transformer relies on the built-in Python parser to generate a concrete syntax tree. It generates an abstract syntax tree from the concrete tree.

The transformer module was created by Greg Stein and Bill Tutt for an experimental Python-to-C compiler. The current version contains a number of modifications and

improvements, but the basic form of the abstract syntax and of the transformer are due to Stein and Tutt.

AST NODES

The compiler.ast module is generated from a text file that describes each node type and its elements. Each node type is represented as a class that inherits from the abstract base class compiler.ast.Node and defines a set of named attributes for child nodes.

class compiler.ast.Node

The Node instances are created automatically by the parser generator. The recommended interface for specific Node instances is to use the public attributes to access child nodes. A public attribute may be bound to a single node or to a sequence of nodes, depending on the Node type. For example, the bases attribute of the Class node, is bound to a list of base class nodes, and the doc attribute is bound to a single node.

Each Node instance has a lineno attribute which may be None. XXX Not sure what the rules are for which nodes will have a useful lineno.

All Node objects offer the following methods:

getChildren()

Returns a flattened list of the child nodes and objects in the order they occur. Specifically, the order of the nodes is the order in which they appear in the Python grammar. Not all of the children are Node instances. The names of functions and classes, for example, are plain strings.

getChildNodes()

Returns a flattened list of the child nodes in the order they occur. This method is like getChildren(), except that it only returns those children that are Node instances.

The While node has three attributes: test, body, and else_. (If the natural name for an attribute is also a Python reserved word, it can't be used as an attribute name. An underscore is appended to the word to make it a legal identifier, hence else_ instead of else.)

The if statement is more complicated because it can include several tests. The If node only defines two attributes: tests and else_. The tests attribute is a sequence of test expression, consequent body pairs. There is one pair for each if/elif clause. The first element of the pair is

the test expression. The second elements is a Stmt node that contains the code to execute if the test is true.

The getChildren() method of If returns a flat list of child nodes. If there are three if/elif clauses and no else clause, then getChildren() will return a list of six elements: the first test expression, the first Stmt, the second text expression, etc.

The following table lists each of the Node subclasses defined in compiler.ast and each of the public attributes available on their instances. The values of most of the attributes are themselves Node instances or sequences of instances. When the value is something other than an instance, the type is noted in the comment. The attributes are listed in the order in which they are returned by getChildren() and getChildNodes().

DEVELOPMENT ENVIRONMENTS:

Most Python implementations (including CPython) include a read–eval–print loop (REPL), permitting them to function as a command line interpreter for which the user enters statements sequentially and receives results immediately.

Other shells, including IDLE and IPython, add further abilities such as auto completion, session state retention and syntax highlighting.

IMPLEMENTATIONS

Reference implementation

CPython is the reference implementation of Python. It is written in C, meeting the C89 standard with several select C99 features. It compiles Python programs into an intermediate bytecode which is then executed by its virtual machine. CPython is distributed with a large standard library written in a mixture of C and native Python. It is available for many platforms, including Windows and most modern Unix-like systems. Platform portability was one of its earliest priorities.

Other implementations

PyPy is a fast, compliant interpreter of Python 2.7 and 3.5. Its just-in-time compiler brings a significant speed improvement over CPython but several libraries written in C cannot be used with it.

Stackless Python is a significant fork of CPython that implements microthreads; it does not use the C memory stack, thus allowing massively concurrent programs. PyPy also has a stackless version.

MicroPython and CircuitPython are Python 3 variants optimized for microcontrollers. This includes Lego Mindstorms EV3.

RustPython is a Python 3 interpreter written in Rust.

Unsupported implementations

Other just-in-time Python compilers have been developed, but are now unsupported:

Google began a project named Unladen Swallow in 2009, with the aim of speeding up the Python interpreter five-fold by using the LLVM, and of improving its multithreading ability to scale to thousands of cores, while ordinary implementations suffer from the global interpreter lock.

Psyco is a just-in-time specialising compiler that integrates with CPython and transforms bytecode to machine code at runtime. The emitted code is specialized for certain data types and is faster than standard Python code.

In 2005, Nokia released a Python interpreter for the Series 60 mobile phones named PyS60. It includes many of the modules from the CPython implementations and some additional modules to integrate with the Symbian operating system. The project has been kept up-to-date to run on all variants of the S60 platform, and several third-party modules are available. The Nokia N900 also supports Python with GTK widget libraries, enabling programs to be written and run on the target device.

Cross-compilers to other languages

There are several compilers to high-level object languages, with either unrestricted Python, a restricted subset of Python, or a language similar to Python as the source language:

- Jython enables the use of the Java class library from a Python program.
- IronPython follows a similar approach in order to run Python programs on the .NET Common Language Runtime.
- The RPython language can be compiled to C, and is used to build the PyPy interpreter of Python.
- Pyjs compiles Python to JavaScript.

- Cython compiles Python to C and C++.
- Numba uses LLVM to compile Python to machine code.
- Pythran compiles Python to C++.
- Somewhat dated Pyrex (latest release in 2010) and Shed Skin (latest release in 2013) compile to C and C++ respectively.
- Google's Grumpy compiles Python to Go.
- MyHDL compiles Python to VHDL.
- Nuitka compiles Python into C++.

PERFORMANCE

A performance comparison of various Python implementations on a non numerical (combinatorial) workload was presented at EuroSciPy '13.

USES

Python has been successfully embedded in many software products as a scripting language, including in finite element method software such as Abaqus, 3D parametric modeler like FreeCAD, 3D animation packages such as 3ds Max, Blender, Cinema 4D, Lightwave, Houdini, Maya, modo, MotionBuilder, Softimage, the visual effects compositor Nuke, 2D imaging programs like GIMP, Inkscape, Scribus and Paint Shop Pro, and musical notation programs like scorewriter and capella. GNU Debugger uses Python as a pretty printer to show complex structures such as C++ containers. Esri promotes Python as the best choice for writing scripts in ArcGIS. It has also been used in several video games, and has been adopted as first of the three available programming languages in Google App Engine, the other two being Java and Go.

Python is commonly used in artificial intelligence projects with the help of libraries like TensorFlow, Keras and Scikit-learn. As a scripting language with modular architecture, simple syntax and rich text processing tools, Python is often used for natural language processing.

Many operating systems include Python as a standard component. It ships with most Linux distributions, AmigaOS 4, FreeBSD (as a package), NetBSD, 42 OpenBSD (as a package) and macOS and can be used from the command line (terminal). Many Linux distributions use installers written in Python: Ubuntu uses the Ubiquity installer, while Red Hat

Linux and Fedora use the Anaconda installer. Gentoo Linux uses Python in its package management system, Portage.

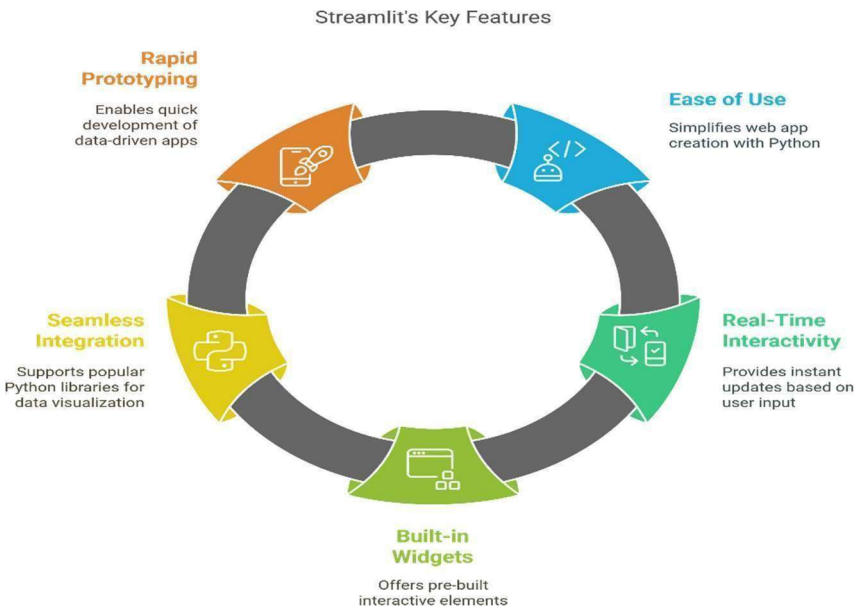
Python is used extensively in the information security industry, including in exploit development.

Most of the Sugar software for the One Laptop per Child XO, now developed at Sugar Labs, is written in Python. The Raspberry Pi single-board computer project has adopted Python as its main user-programming language.

LibreOffice includes Python, and intends to replace Java with Python. Its Python Scripting Provider is a core feature since Version 4.0 from 7 February 2013.

STREAMLIT

Streamlit is an open-source Python framework used to build interactive web applications easily, especially for data science and machine learning projects. It lets developers turn Python scripts into shareable web apps without needing web development skills like HTML, CSS, or JavaScript. With Streamlit, you can display data, charts, and models directly in the browser and update the app instantly when the code changes.



Library features

- Very easy to use with simple Python code

- No need for frontend (HTML/CSS/JS) knowledge
- Supports charts, tables, and data visualization
- Works well with Pandas, NumPy, Matplotlib, and ML libraries
- Real-time updates when code is saved
- Built-in widgets like buttons, sliders, and text inputs
- Good for rapid prototyping and sharing apps

CHAPTER 5

SYSTEM DESIGN

5.1 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.

4. Provide a formal basis for understanding the modeling language.

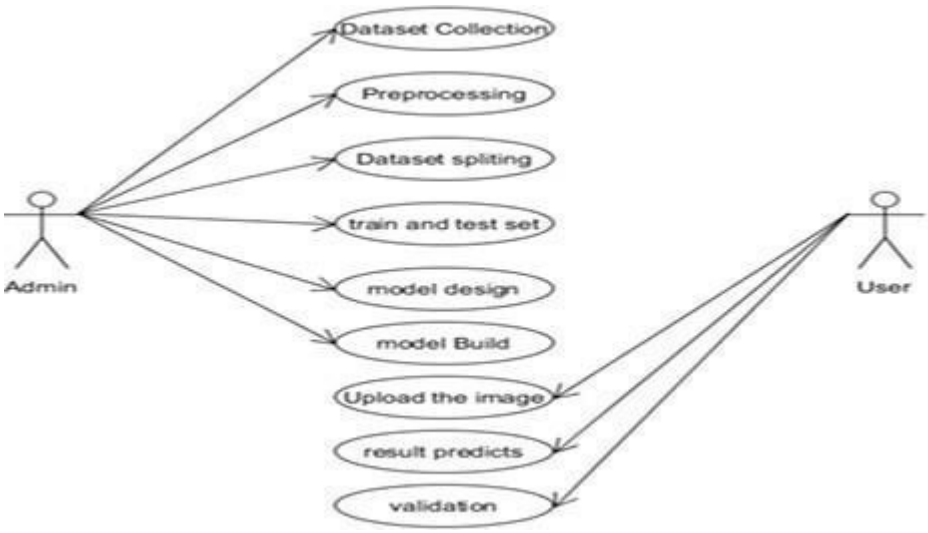
5. Encourage the growth of OO tools market.

6. Support higher level development concepts such as collaborations, frameworks, patterns and components.

7. Integrate best practices.

5.2 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



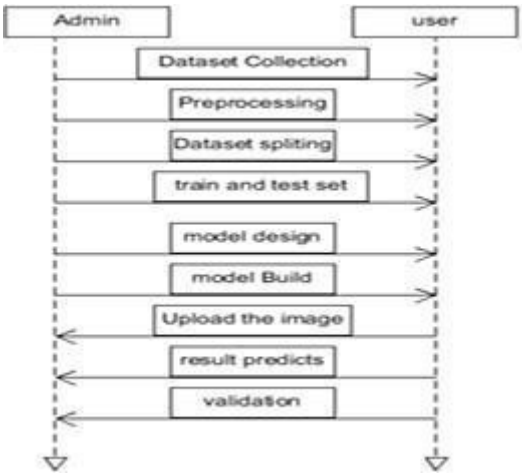
CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the

system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

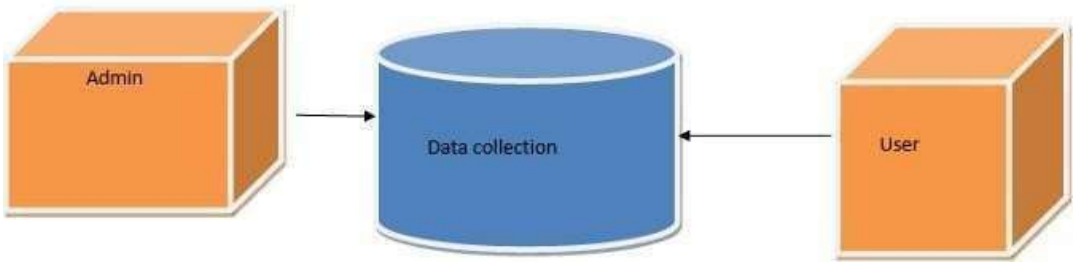
5.3 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.



DEPLOYMENT

Component diagrams are used to describe the components and deployment diagrams shows how they are deployed in hardware. UML is mainly designed to focus on the software artifacts of a system. However, these two diagrams are special diagrams used to focus on software and hardware components.



DATA FLOW DIAGRAM:

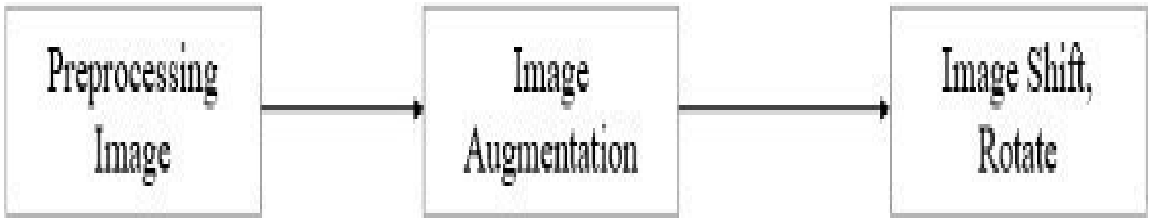
The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.

1. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
2. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output. 4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

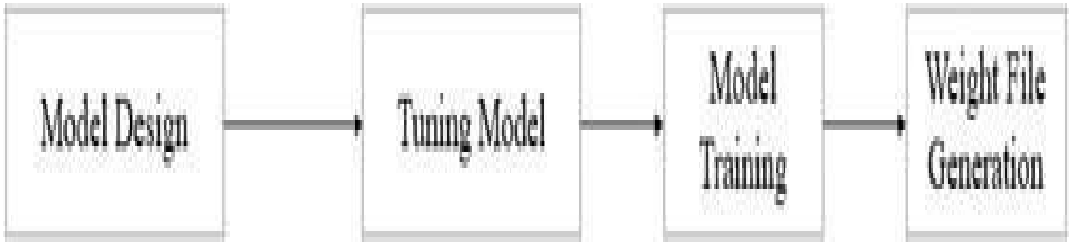
DFD DIAGRAM:

LEVEL 0:

LEVEL 1:


Preprocessing Image
Image Augmentation
Image Shift, Rotate

LEVEL 2:


Model Design
Tuning Model
Model Training
Weight File Generation

CHAPTER 6

SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by 51 successfully unit testing, the combinaation of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least 52 its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

6.1 Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

6.2 Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results

All the test cases mentioned above passed successfully. No defects encountered.

6.3 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results

All the test cases mentioned above passed successfully. No defects encountered.

CHAPTER 7

SOFTWARE IMPLEMENTATION

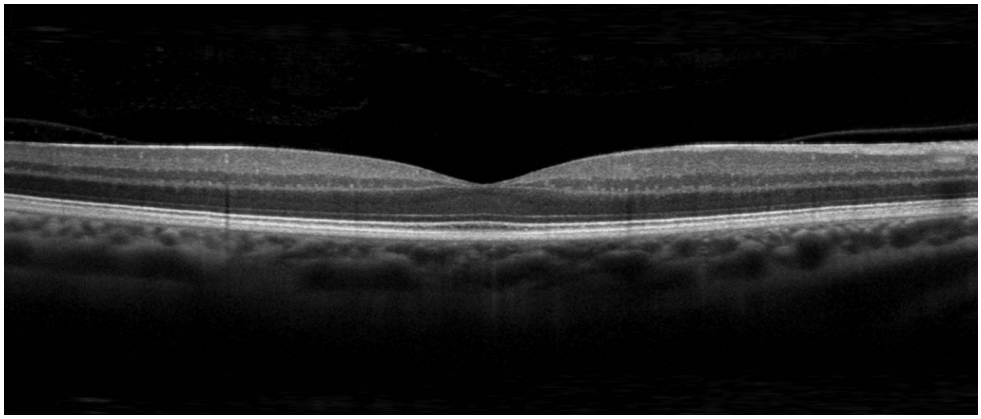
7.1 SOURCE CODE

```
import streamlit as st

import numpy as np

from tensorflow.keras.models import load_model

from tensorflow.keras.preprocessing import image

from PIL import Image

import os


# ========================

# CONFIG

# ========================

os.environ["TF_CPP_MIN_LOG_LEVEL"] = "2"


st.set_page_config(

    page_title="OCT Image Classification",

    layout="centered"

)


# ========================

# TITLE
```

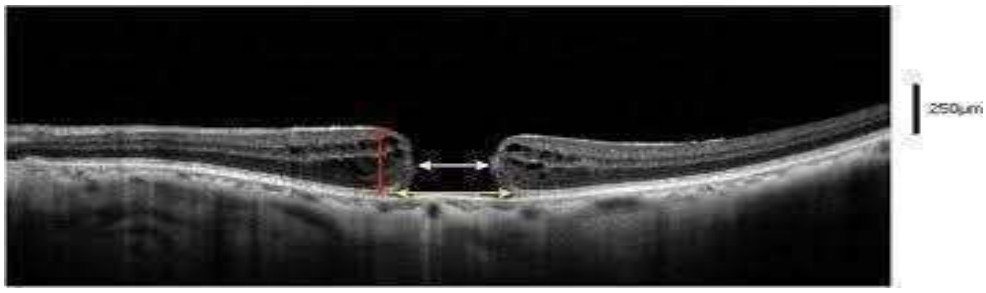
```
# ========================
st.title("OCT Image Classification")
st.write("Upload an OCT scan image and the model will predict its class.")


# ========================
# LOAD MODEL
# ========================
@st.cache_resource
def load_oct_model():
    return load_model("oct_model_python312.h5")


model = load_oct_model()
# ========================
# CLASS NAMES
# ı MUST MATCH MODEL OUTPUT ORDER #
========================
CLASS_NAMES=["AMD","CNV","CSR",  "DME","DR","DRUSEN","MH","NORMAL",]


# ========================
# CHECK MODEL OUTPUT
# ========================
NUM_CLASSES =  model.output_shape[-1]


if NUM_CLASSES != len(CLASS_NAMES):
```

```
    st.error(
        f"Model outputs {NUM_CLASSES} classes, "
        f"but CLASS_NAMES has {len(CLASS_NAMES)} labels."
    )
    st.stop()

# ========================
# IMAGE UPLOAD
# ========================
uploaded_file = st.file_uploader(
    "Choose an OCT image",
    type=["jpg", "png", "jpeg"]
)

if uploaded_file is not None:
    # ----------------------------
    # LOAD IMAGE (FORCE RGB)
    # --------------------------
    img = Image.open(uploaded_file).convert("RGB")
    st.image(img, caption="Uploaded Image", use_column_width=True)
```

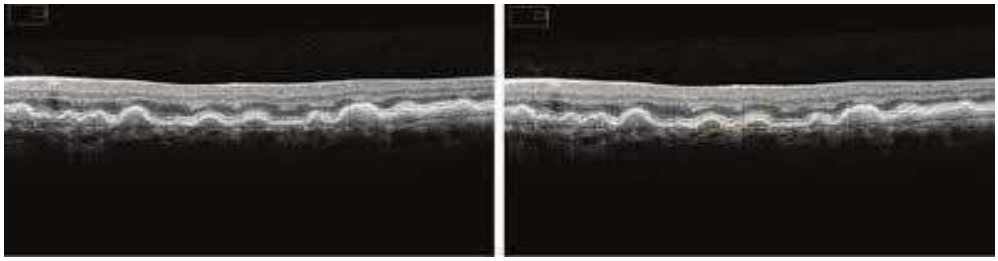
INPUT IMAGES



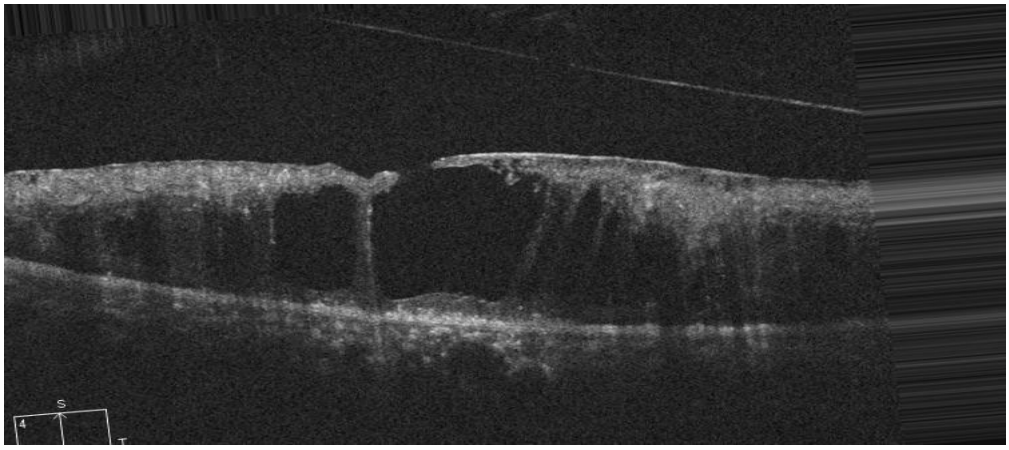
NORMAL IMAGE



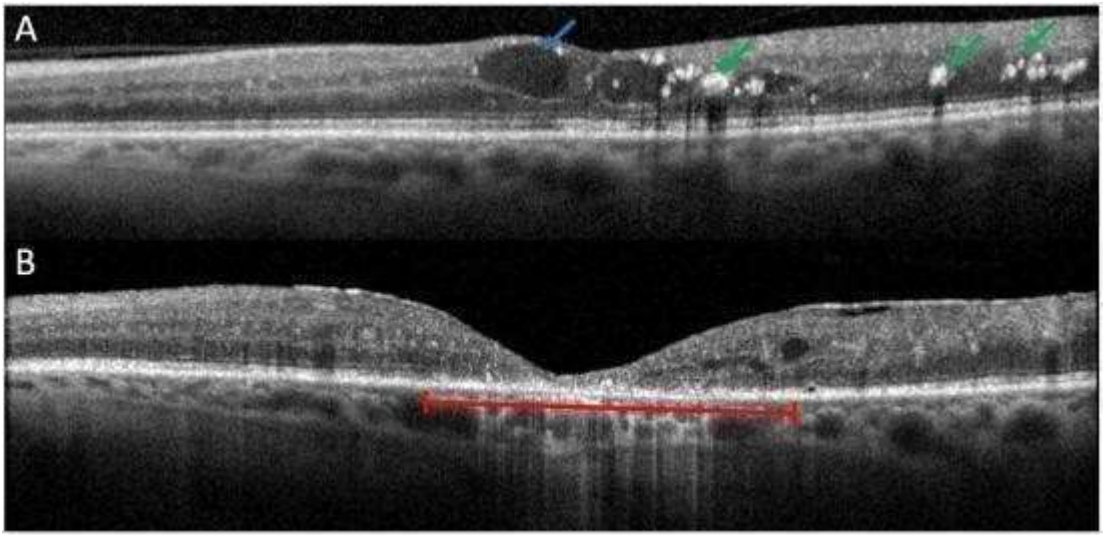
MH IMAGE



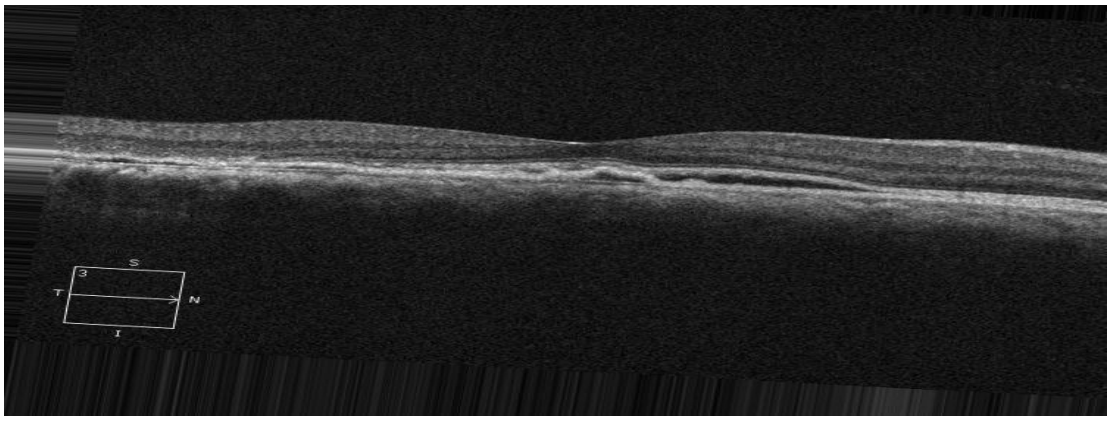
DRUSEN IMAGE



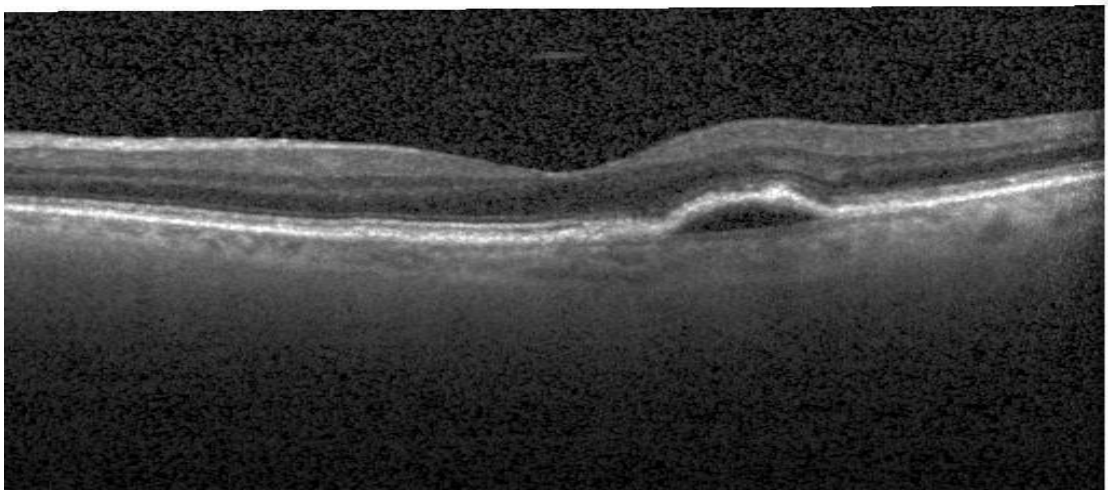
DR IMAGE



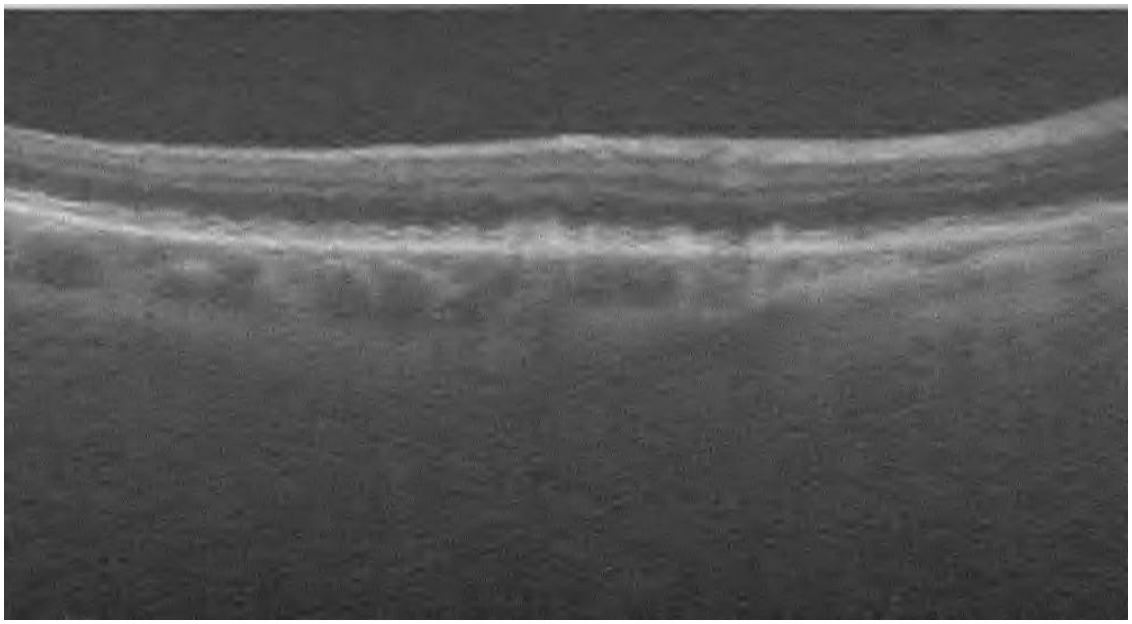
DME IMAGE



CSR IMAGE

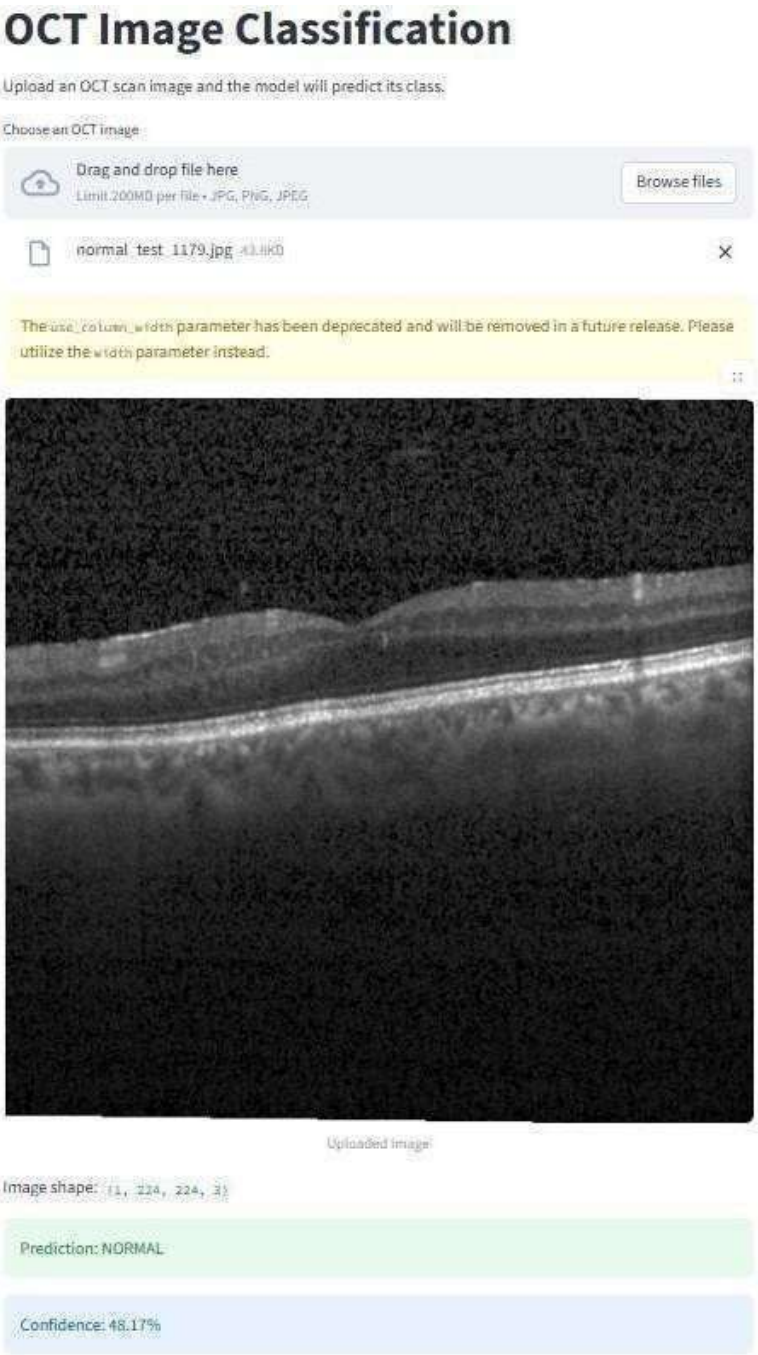


CNV IMAGE



AMD IMAGE

7.2 SCREEN LAYOUTS



OCT Image Classification
Upload an OCT scan image and the model will predict its class.
Choose an OCT image
Drag and drop file here
Limit 200MB per file • JPG, PNG, JPEG
Browse files
mh_train_2979.jpg 57.2KB
The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.
Uploaded Image
Image shape: (1, 224, 224, 3)
Prediction: MH
Confidence: 100.00%

OCT Image Classification

Upload an OCT scan image and the model will predict its class.

Choose an OCT image

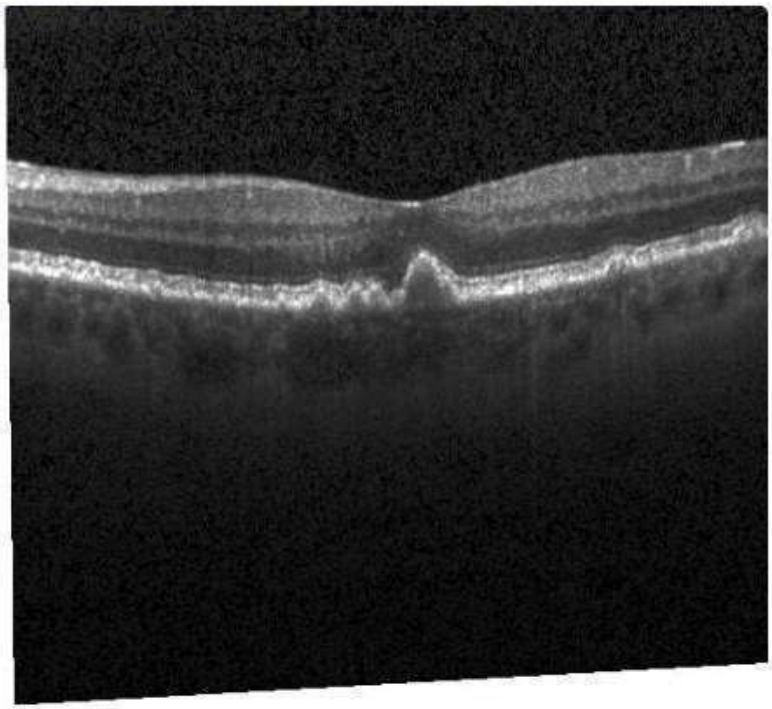
Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files

drusen_train_3263.jpg 24.6KB

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



Uploaded Image

Image shape: (1, 224, 224, 3)

Prediction: DRUSEN

Confidence: 98.71%

OCT Image Classification

Upload an OCT scan image and the model will predict its class.

Choose an OCT image

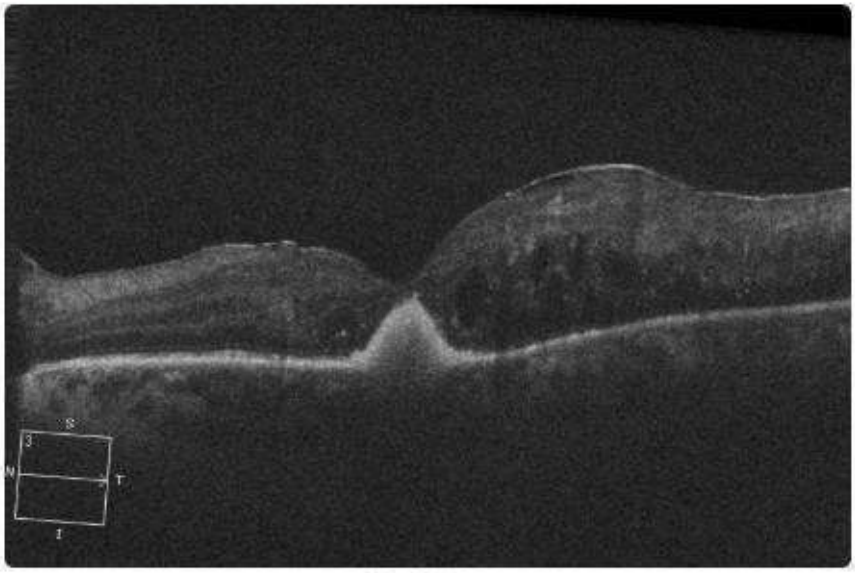
Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files

dr_train_3161.jpg 54.8KB

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



Uploaded Image

Image shape: (1, 224, 224, 3)

Prediction: DR

Confidence: 99.96%

OCT Image Classification

Upload an OCT scan image and the model will predict its class.

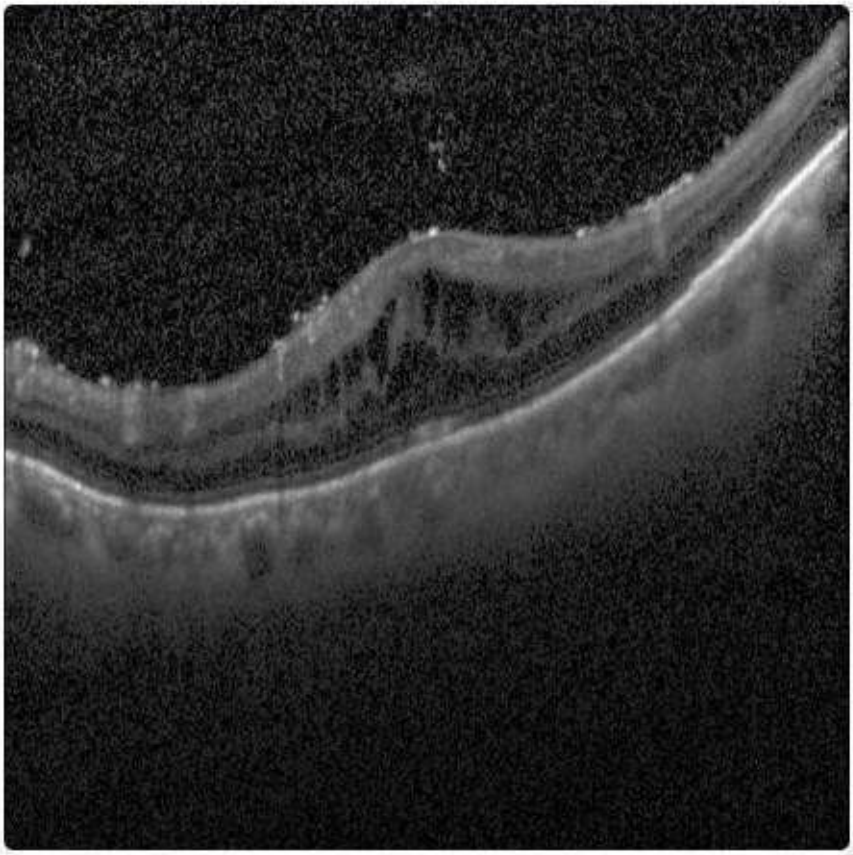
Choose an OCT image

Drag and drop file here
Limit 200MB per file • JPG, PNG, JPEG

Browse files

dme_train_3228.jpg 97.4KB

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



Uploaded Image

Image shape: (1, 224, 224, 3)

Prediction: DME

Confidence: 100.00%

OCT Image Classification

Upload an OCT scan image and the model will predict its class.

Choose an OCT image

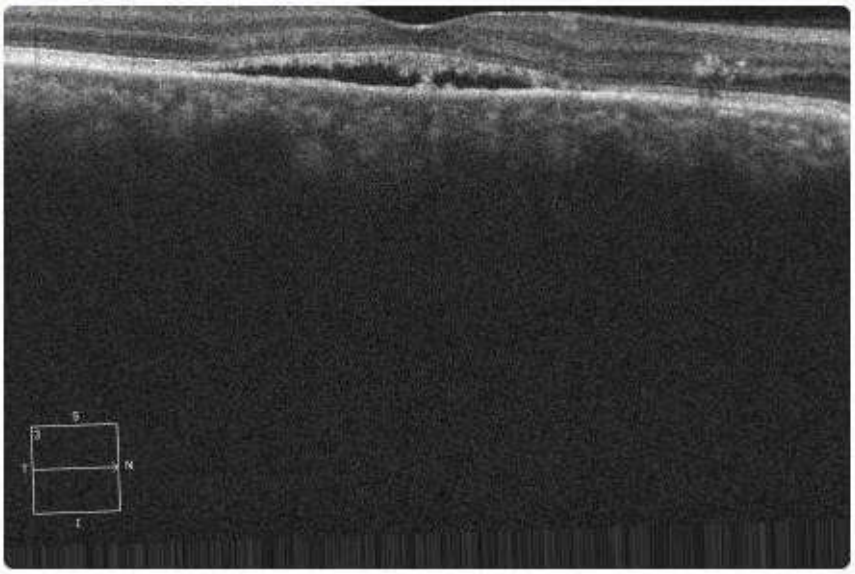
Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files

csr_train_3148.jpg 86.3KB

The `use_column_width` parameter has been deprecated and will be removed in a future release. Please utilize the `width` parameter instead.



Uploaded Image

Image shape: `(1, 224, 224, 3)`

Prediction: CSR

Confidence: 99.99%

OCT Image Classification

Upload an OCT scan image and the model will predict its class.

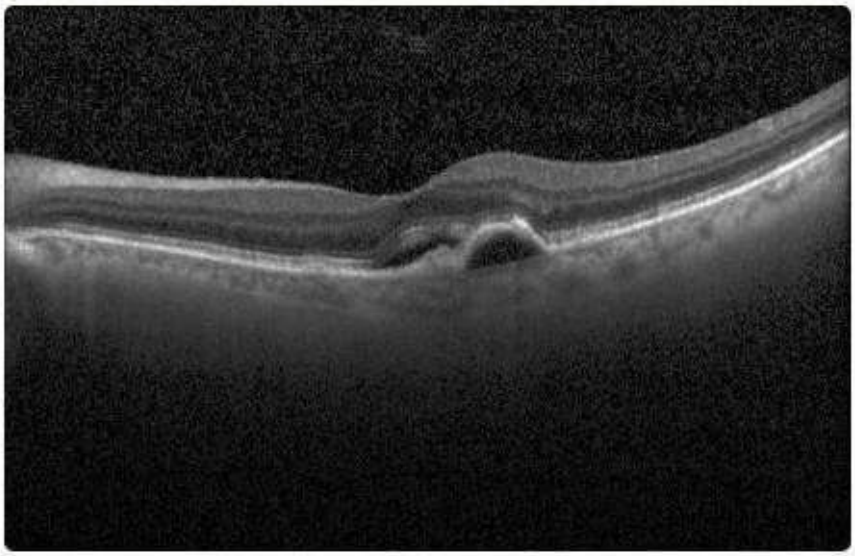
Choose an OCT image

Drag and drop file here
Limit 200MB per file • JPG, PNG, JPEG

Browse files

cnv_train_3270.jpg 61.7KB

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



Uploaded image

Image shape: (1, 224, 224, 3)

Prediction: CNV

Confidence: 93.28%

OCT Image Classification

Upload an OCT scan image and the model will predict its class.

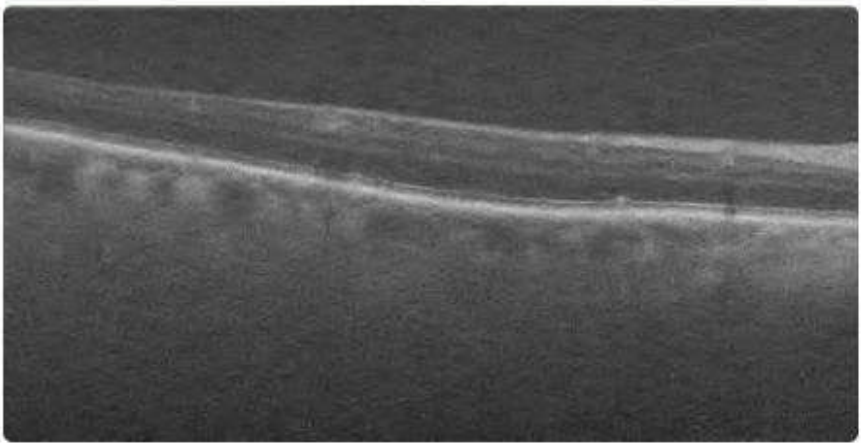
Choose an OCT image

Drag and drop file here
Limit 200MB per file • JPG, PNG, JPEG

Browse files

amd_train_3285.jpg 66.5KB

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



Uploaded image

Image shape: (1, 224, 224, 3)

Prediction: AMD

Confidence: 100.00%

CONCLUSION

In this project, the application of the neural network models with effective feature reuse features based on the retinal OCT datasets in the diagnosis of AMD, DME was tested and evaluated. The optimization strategies of different neural network models, the effects of different neural network models on final diagnostic recognition and the adaptability of different neural network models to different datasets were studied. Two publicly available OCT datasets were used to evaluate the performance of the neural networks we selected and the impact of differences among datasets. Five neural network structures were tested and evaluated (DenseNet121, ResNet50, DPN92, ResNext101, CliqueNet) and VGG16, VGG19, inception-V3 neural networks, and SVM methods were added to improve experimental comparisons. Accuracy, recall, precision, AUC were used as evaluation criterias to show the ROC curve and confusion matrix of each neural network. After the testing, CliqueNet achieves the best score, with a 0.99/1.00 three types average AUC, a 0.992 three-types average recall rate, 0.985 three-types average precision, and a 0.990 of accuracy rate on dataset 1 and a 0.99/0.99 of three types average AUC, a three-types average recall of 0.988, an average precision of 0.979, and an accuracy of 0.986. The effects achieved by CliqueNet on different datasets are similar. After analysis and comparison, two neural network models CliqueNet and DPN were obtained, which have better effects on different datasets. CliqueNet and DPN can be applied to actual clinical diagnosis. CliqueNet training is complex but result is better. Although DPN is not as effective as CliqueNet on dataset 2, the training method is easy, showing that CliqueNet can better help doctors prevent misdiagnosis and missed diagnosis. We believe that when dealing with 55 medical datasets with less data, some neural network models with effective feature of reuse can be taken into account, but also, data amplification through data enhancement and generative adversarial networks (GAN) can be considered.

FUTURE ENHANCEMENT

In the future, the proposed system for classification and detection of OCT retinal images using deep learning can be further improved in several ways. The performance of the model can be enhanced by training it on a larger and more diverse dataset, which will increase accuracy and robustness. Advanced deep learning architectures can be implemented to improve feature extraction and classification of retinal diseases.

The system can also be extended to support multi-disease classification, making it more suitable for real-world clinical use. Real-time detection capability can be added to assist ophthalmologists during diagnosis. Additionally, integrating explainable AI techniques can help visualize affected regions in OCT images, improving trust and interpretability.

The model can be deployed on cloud or mobile platforms to enable remote screening and accessibility in rural areas. Overall, these enhancements will make the system more efficient, accurate, and practical for healthcare applications.

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